

GPS + RTK System Configuration & Setup Guide

Trimble BX982 + Digi TransPort WR21

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1. System Overview

This document describes the full configuration for a GPS + RTK (Real Time Kinematic) positioning system installed on a robot platform, consisting of:

- Trimble BX982 GNSS Receiver (Serial: 5311K89701, IP: 192.168.1.132) — dual-antenna RTK receiver providing position, heading, and attitude
- Digi TransPort WR21 Router (SN: 304861, IP: 192.168.1.2) — cellular router providing internet connectivity for RTK corrections via SIM card
- AG25 GNSS Antenna x2 — Antenna 1 (front-left) and Antenna 2 (rear-right) on the robot
- NTRIP RTK correction service via vrs.axis-gps.com — contact AXIS GPS at 04-6419995

2. Hardware Setup & Connections

2.1 Trimble BX982 GNSS Receiver

The yellow Trimble BX982 receiver box is the central unit. It is accessed via web browser at 192.168.1.132:444.

Parameter	Value
Model	Trimble BX982 GNSS Receiver
Serial Number	5311K89701
Firmware Version	5.11 (2015-12-16)
IP Address	192.168.1.132
Web UI Port	444 (http://192.168.1.132:444)
System Name	Trimble



Fig 1: Trimble BX982 (yellow) with Digi WR21 (grey, on top) — front view showing LAN1, Serial, power and WWAN antenna ports



Fig 2: Rear view of Trimble BX982 — showing PORT4 (TO BX Data), CAN, Data/Power, Pos Ant1 and Pos Ant2 connectors

Trimble BX982 Port Connections

Parameter	Value
LAN 1 (Ethernet)	Connected to Digi WR21 LAN 1 port
Pos Ant1	AG25 GNSS Antenna 1 — Front-Left of robot
Pos Ant2	AG25 GNSS Antenna 2 — Rear-Right of robot
Data/Power	Main power and data connector
PORT4 (TO BX Data)	Serial data output to other systems
CAN	CAN bus connection
Serial B	Connected to Digi WR21 Serial B port

Note: The LAN1 port on the Trimble connects to LAN1 on the Digi WR21. The Digi LAN0 port connects to the onboard computer. This keeps the GPS network separate from the computer network on the Digi side.

2.2 Digi TransPort WR21 Router

The Digi WR21 router sits on top of the Trimble BX982. It provides cellular (4G/LTE) internet via a SIM card, which is used to receive RTK correction data.

Parameter	Value
Model	Digi TransPort WR21
Serial Number	304861
IP Address	192.168.1.2
Web UI	http://192.168.1.2/login.asp
Login Username	username
Login Password	password



Fig 3: Digi TransPort WR21 — front face showing SIM 1, SIM 2 slots, POWER, SERVICE, WWAN LEDs, and USB port

Digi WR21 Port Connections

Parameter	Value
LAN 0 (Ethernet)	Connected to the onboard computer
LAN 1 (Ethernet)	Connected to Trimble BX982 LAN1 port
SIM 1 / SIM 2	Insert cellular SIM card (see Section 2.3)
WWAN Primary antenna	Cellular antenna
WWAN Secondary antenna	Cellular diversity antenna
Serial B	Connected to Trimble BX982 Serial B

2.3 SIM Card Installation

The SIM card must be inserted for cellular connectivity and RTK correction data to function.

1. Power off the Digi WR21.
2. Locate the SIM 1 slot on the front-left face of the unit (visible in Fig 3 above).
3. Insert a data-enabled SIM card. The WR21 supports standard SIM form factor.
4. Power the router back on.
5. Verify mobile connectivity: in the Digi web UI, go to Management > Network Status and confirm the WWAN (mobile) interface shows as connected.
6. The WWAN LED on the front panel should be lit green when connected.

Note: Contact your cellular provider for APN settings. For help with RTK service activation call AXIS GPS at 04-6419995.

3. Antenna Placement on Robot

Two Trimble AG25 GNSS antennas are mounted on the robot. Their positions relative to each other determine the heading accuracy.

Parameter	Value
Antenna 1 (Pos Ant1)	Front-Left of robot
Antenna 2 (Pos Ant2)	Rear-Right of robot
Delta X (lateral separation)	1.2 meters
Delta Y (longitudinal separation)	1.7 meters
Baseline length	$\sim 2.07 \text{ m } (\sqrt{1.2^2+1.7^2})$
Heading adjustment configured	46.300°

Top-View Diagram — Robot Antenna Layout

FRONT-LEFT	FRONT	FRONT-RIGHT
Ant 1 (Pos Ant1) Front-Left	ROBOT BODY Delta X = 1.2 m Delta Y = 1.7 m	
REAR-LEFT	REAR	Ant 2 (Pos Ant2) Rear-Right

Note: Antenna 1 connects to the Pos Ant1 connector on the BX982; Antenna 2 connects to Pos Ant2. The heading adjustment of 46.300° in the Vector Configuration compensates for the diagonal baseline angle relative to the robot's forward axis. If antennas are moved, this value must be recalculated and updated.

Heading Adjustment Calculation

The heading correction offset is the angle of the antenna baseline relative to the robot forward (Y) axis:

$$\text{Offset angle} = \arctan(\text{Delta X} / \text{Delta Y}) = \arctan(1.2 / 1.7) \approx 35.2^\circ$$

Note: The configured value of 46.300° may include additional mechanical offset from the robot frame. Verify by driving the robot straight and comparing GPS heading to actual heading, then fine-tune as needed.

4. Trimble Receiver Configuration

4.1 General / Summary Settings

Navigate to: Receiver Configuration > Summary

Parameter	Value
Elevation Mask	10°
PDOP Mask	7
Horizontal Precision	0.30 m
Vertical Precision	0.30 m
Clock Steering	Disabled
Everest™ Multipath Mitigation	Enabled
Signal Tracking Bandwidth	Wide
RTK Mode	Low Latency
Motion	Kinematic
CMR Input Filter	Disabled

4.2 Position Antenna (Ant1 — Front-Left)

Navigate to: Receiver Configuration > Position Antenna

Parameter	Value
Antenna Type	Unknown External
RINEX Name	Unknown External
Antenna Measurement Method	Antenna Phase Center
Antenna Height	2.010 m
Apply Antenna Correction to	RTCM V3 (checked)

4.3 Vector Antenna (Ant2 — Rear-Right)

Navigate to: Receiver Configuration > Vector Antenna

Parameter	Value
Antenna Type	AG25 GNSS
RINEX Name	TRMAG25 NONE

Antenna Measurement Method	Antenna Phase Center
Antenna Height	2.010 m
Vector Antenna ID	266

4.4 Vector / Heading Configuration

Navigate to: Receiver Configuration > Vector (Tracking)

Heading

Parameter	Value
Heading Adjustment	46.300°
Direction	Normal
Apply Adjustment To	All heading outputs

Attitude

Parameter	Value
Slope Adjustment	0.000°
Calibration Options	Unchecked
Direction	Normal
Orientation	Unspecified

Minimum Solution Type

Parameter	Value
Minimum Solution Type	RTK Fixed

4.5 Position Settings

Navigate to: Receiver Configuration > Position

Parameter	Value
PDOP Mask	7
RTK Mode	Low Latency
Autonomous/Differential Engine	Kalman / SBAS
Signal Tracking Bandwidth	Wide

Receiver Motion (Dynamic Model)	Kinematic
DGNSS Age of Correction (GPS)	60 Sec.
ITRF Realization (2008) Epoch	Current

4.6 Reference Station

Navigate to: Receiver Configuration > Reference Station

Parameter	Value
Reference Latitude	32°00'11.28730"N
Reference Longitude	34°48'56.25285"E
Reference Height	50.687 m
Station Name	NULLNAME
RTCM 2.x ID	0
RTCM 3.x ID	0
CMR ID	0

5. I/O Port Configuration

Navigate to: I/O Configuration > Port Summary

Type	Port / Address	Input	Output
TCP/IP	667	-	OmniDATA (Trimble)
TCP/IP	5018	-	NMEA-GGA (1Hz)
TCP/IP	28001	-	NMEA-GGA (10Hz), NMEA-HDT (10Hz)
TCP/IP	28002	-	GSOFF (10Hz)
TCP/IP	28003	-	NMEA-GGA (1Hz), NMEA-VHD (1Hz)
NTRIP Client 1	vrs.axis-gps.com:2102/VRS-RTX	RTCMv3	NMEA-GGA (10 Sec.)
Serial COM1	115K-8N1	-	NMEA-GGA (10Hz), NMEA-HDT (10Hz)
Serial COM2	115K-8N1	-	-
Serial COM4	115K-8N1	-	NMEA-GGA (10Hz), NMEA-HDT (2Hz)
CAN	CAN 1	-	-

Note: Green-highlighted rows are active/connected ports. NTRIP Client 1 receives RTK corrections. TCP/IP port 28002 outputs GSOFF at 10Hz used by downstream systems.

6. RTK Correction Service (NTRIP)

6.1 Service Details

Parameter	Value
NTRIP Caster Host	vrs.axis-gps.com
Port	2102
Mountpoint	VRS-RTX
Protocol	RTCMv3
GGA Feedback Rate	Every 10 seconds (for VRS positioning)
Service Contact	04-6419995 (AXIS GPS)

6.2 Activation Steps

7. Call AXIS GPS at 04-6419995 to activate service and obtain NTRIP username/password.
8. Log in to Trimble web UI at <http://192.168.1.132:444>.
9. Navigate to I/O Configuration > Port Configuration > NTRIP Client 1.
10. Enter caster address: vrs.axis-gps.com:2102/VRS-RTX
11. Enter credentials provided by AXIS GPS.
12. Set Input = RTCMv3, Output = NMEA-GGA (10 Sec.).
13. Click OK. The NTRIP Client 1 row should turn green in Port Summary.

Note: The Digi WR21 must have an active SIM with data enabled before the NTRIP connection will work.

7. Digi TransPort WR21 Configuration

7.1 Network Layout

Parameter	Value
Digi WR21 LAN 0	192.168.1.2 — to onboard computer
Digi WR21 LAN 1	192.168.1.x — to Trimble BX982 (192.168.1.132)
Trimble BX982 LAN1	192.168.1.132
Subnet	192.168.1.0/24

Note: LAN 0 faces the computer network; LAN 1 faces the GPS/Trimble network. The Digi router bridges cellular internet to the LAN so the Trimble can reach the NTRIP caster.

7.2 Mobile Interface Configuration

Navigate to: Configuration > Network > Interfaces > Mobile

14. Ensure SIM is inserted in SIM 1 slot.
15. Enable the Mobile / WWAN interface.
16. Enter APN, username and password from your cellular provider.
17. Save and reboot the router.
18. Verify: Management > Network Status — WWAN should show as connected with an IP address.
19. Front panel WWAN LED should be green.

8. System Startup Checklist

#	Step
1	Confirm SIM card is seated in Digi WR21 SIM 1 slot
2	Power on Digi WR21 — verify WWAN LED turns green (cellular connected)
3	Check Digi LAN 0 → computer cable and LAN 1 → Trimble BX982 LAN1 cable are connected
4	Power on Trimble BX982 receiver
5	Wait for satellite acquisition — check Receiver Status > Satellites in Trimble web UI
6	Verify NTRIP Client 1 row is green in I/O Configuration > Port Summary
7	Confirm RTK Fixed solution achieved (check Receiver Status)
8	Verify heading output active on COM1/COM4 and TCP port 28001/28002
9	Confirm heading adjustment is 46.300° in Receiver Configuration > Vector

9. Troubleshooting

No RTK Fix / NTRIP Client 1 red

- Verify SIM card is inserted and WWAN LED is green on Digi WR21.
- Check Digi web UI: Management > Connections — confirm NTRIP connection is active.
- Confirm NTRIP credentials are correct — call AXIS GPS at 04-6419995.
- Check LAN 1 cable between Digi and Trimble BX982 LAN1 port.

No Satellites / Poor Satellite Count

- Check both AG25 antenna cables are firmly connected at Pos Ant1 and Pos Ant2.
- Ensure both antennas have clear sky view above 10° elevation.
- Confirm Elevation Mask = 10° in Receiver Configuration.

Wrong Heading Output

- Verify Heading Adjustment = 46.300° in Receiver Configuration > Vector.
- Confirm Antenna 1 is at front-left and Antenna 2 is at rear-right (not swapped).
- Check Delta X = 1.2 m, Delta Y = 1.7 m matches physical installation.
- Minimum Solution Type must be RTK Fixed for heading to be valid.

Cannot Access Trimble Web UI

- Confirm device is on 192.168.1.x subnet via LAN 1 of Digi WR21.
- Navigate to <http://192.168.1.132:444> (port 444, not 443).

- Ping 192.168.1.132 to verify network connectivity.

Cannot Access Digi Web UI

- Navigate to <http://192.168.1.2/login.asp> via LAN 0 (computer port).
- Username: username / Password: password